

RESEARCH PROJECT BRIEF

Life in the Stars — Could Plants Exist Beyond Earth?

Exploring plant life possibilities across our Solar System and the Milky Way

Project Title	Life in the Stars — Could Plants Exist Beyond Earth?
Topic	The possibility of plant-like life across our Solar System and the Milky Way
Project Type	Independent Research & Web Development
Total Marks	100 Marks
Deliverables	2 (Research Report + Live Website)
Timeline	Research Phase (Week 1–2) Build Phase (Week 3–4)

Project Overview

For thousands of years humans have looked up and wondered — is anything else out there? This project takes that question seriously. You will research the real science behind plant-like life beyond Earth: what conditions plants need to survive, which worlds in our Solar System could support them, what alien plants might look like under a different sun, and how scientists are searching the Milky Way for signs of photosynthetic life right now.

This project has two phases:

- Deliverable 1 — Deep Research: Investigate the science of plant life and its potential across the cosmos.
- Deliverable 2 — Website Build: Build a bold, immersive website that presents your research like a space documentary.

Learning Objectives

- Research a complex scientific topic using credible, multi-source references
- Understand and explain real astrobiology, planetary science, and botany concepts
- Translate dense scientific content into clear, engaging writing for a general audience
- Design and build an immersive, visually compelling website from scratch
- Connect science, creativity, and design thinking in a single cohesive project

Instructions

1. All written research must be in your own words — summarise, paraphrase, and cite your sources.
2. All website code must be written by you. AI-generated code submitted as your own is not permitted.
3. Your website must be fully functional — all links, sections, and interactive elements must work.
4. Include clear evidence of your research process (notes, source list, screenshots).
5. Submit both deliverables by the end of Week 4.

DELIVERABLE 1: Research Report

Total: 50 Marks

Your first deliverable is a structured research report exploring the possibility of plant-like life beyond Earth. You will investigate real science — planetary conditions, extremophile biology, astrobiology, and the search for life across the Milky Way — and present your findings in a well-organised, original written report.

1.1 Introduction — Why Plants, Why Space?

Section 1.1 — Introduction

10 Marks

Write a 1–2 page introduction to the topic. Your introduction must address the following:

- What do plants actually need to survive? (light, water, atmosphere, soil chemistry — in your own words)
- Why is this question relevant? What does searching for plant-like life tell us about our place in the universe?
- What is astrobiology and how does it guide the search for life?
- A brief preview of the worlds and ideas you will explore in this report
- Your own reflection: what draws you to this topic and what you expect to discover

Evidence required: Written introduction (350–500 words) + at least 1 supporting image or diagram with caption.

1.2 Five Worlds — Could Plants Survive There?

Section 1.2 — Planetary Research

20 Marks

Research and write a dedicated section for each of the following five worlds. For each one, use the structure below:

Per World	Basic facts Atmosphere & conditions Could plants survive? (yes/no/maybe — justify it) What type of plant could theoretically exist? 1 image with caption
World 1	Mars — The Red Planet
World 2	Europa — Jupiter's ice moon
World 3	Titan — Saturn's moon with thick atmosphere
World 4	Kepler-442b — An Earth-like exoplanet
World 5	Proxima Centauri b — Nearest star's planet

Minimum per world: 200–300 words + 1 image with caption. Total for Section 1.2: approximately 1,200–1,500 words.

1.3 What Would Alien Plants Look Like?

Section 1.3 — Speculative Science

10 Marks

This section blends real science with informed speculation. Based on your research, write an extended analysis (400–600 words) exploring:

- How would plants adapt to a dim red dwarf star vs a bright blue star?
- What colour might alien plants be, and why? (Hint: research chlorophyll alternatives)
- How could plants survive without liquid water — alternative solvents, underground roots, or dormancy?
- Could plants survive on a tidally locked planet where one side is always day?
- Your conclusion: what does this tell us about how life finds a way?

Evidence required: Extended write-up + minimum 2 images/diagrams + at least 3 cited sources.

1.4 The Search — How Scientists Look for Plant Life

Section 1.4 — Astrobiology Methods

5 Marks

Write a focused section (200–300 words) explaining how scientists actually search for signs of plant-like life in the universe:

- What is the "vegetation red edge" and why does it matter?
- How do space telescopes like James Webb detect biosignatures?
- What gases in an atmosphere would hint at photosynthesis? (oxygen, methane, CO₂)
- Name one real current or planned mission that could find signs of plant life

Evidence required: Written section + 1 supporting image with caption + 2 cited sources.

1.5 Reflection & Source List

Section 1.5 — Reflection & References

5 Marks

Reflection (150–200 words)

- What surprised or challenged you most during your research?
- If plants were found on another world, what would that mean for humanity?
- What connection do you see between this topic and purpose, faith, or meaning?

Source List

- Minimum 6 credible sources (NASA, ESA, science journals, documentaries, university sites)
- Use a consistent citation format (APA or Harvard)
- Mark which sources were used for which section

DELIVERABLE 2: The Website

Total: 50 Marks

Your second deliverable is a fully built, immersive website that presents your research in a way that feels like exploring space. The website is not a copy of your report — it is a visual, interactive experience of it. Think deep space documentary meets nature encyclopedia.

2.1 Website Structure & Pages

Section 2.1 — Structure

10 Marks

Page / Section	Content Required	Status
Home / Landing	Bold hero with starfield or space imagery, project title, short hook intro, navigation	Required
Why Plants in Space?	Visual intro section — what plants need, what astrobiology is	Required
The Five Worlds	One card/panel per world with image, facts, and plant survival verdict	Required
Alien Plant Design	Creative section — what alien plants might look like with visual concepts	Required
The Search	How scientists detect life — telescope visuals, biosignature explainer	Required
Reflection & Sources	Personal reflection + formatted reference list	Required

2.2 Design & Visual Quality

Section 2.2 — Design

15 Marks

The website must feel like space. It should be dark, atmospheric, and immersive — not a plain document. The following design standards apply:

- Dark colour palette — deep space blacks, navy blues, with pops of green or purple
- Full-width hero section with a compelling headline and a space/planet background image
- Each of the 5 worlds must have its own distinct visual card with a planet image
- Consistent typography — 1 to 2 fonts that feel modern and readable
- Use high-quality space imagery (NASA image library is free to use)
- Smooth layout — no broken elements, overflowing text, or missing images
- Mobile-responsive OR clearly optimised for desktop (state which in your README)

Evidence required: Screenshots of every page/section submitted alongside your code files.

2.3 Technical Build Quality

Section 2.3 — Code Quality

15 Marks

Your website must be built with HTML, CSS, and at least one JavaScript feature. Requirements:

- Clean, readable, well-indented code with comments where helpful
- CSS must be in a separate stylesheet — no inline styles for layout
- All navigation links must work and lead to the correct section or page
- At least ONE JavaScript feature implemented (e.g., planet cards that flip, star animation, scroll reveal, image lightbox, tab switching between worlds)
- All images must have descriptive alt text
- Website must open from index.html with no broken paths

Evidence required: Full project folder submitted (HTML + CSS + JS + images + README.txt).

2.4 Content Accuracy & Depth

Section 2.4 — Content on Website

10 Marks

The content on your website must accurately reflect your research from Deliverable 1. Markers will check that:

- Each world section contains accurate, research-backed information — not vague or made-up claims
- The alien plant design section is grounded in real science (colour, light, chemistry)
- The biosignature / search section explains real methods used by scientists today
- All content is written in your own words — not copied from your report
- The reflection section is genuine and personal

Marking Summary

#	Component	Marks	Weight
1.1	Introduction — Why Plants, Why Space?	10	10%
1.2	Five Worlds — Could Plants Survive There?	20	20%
1.3	What Would Alien Plants Look Like?	10	10%
1.4	The Search — How Scientists Look for Plant Life	5	5%
1.5	Reflection & Source List	5	5%
2.1	Website Structure & Pages	10	10%
2.2	Design & Visual Quality	15	15%
2.3	Technical Build Quality	15	15%
2.4	Content Accuracy & Depth	10	10%
	TOTAL	100	100%

Submission Checklist

Tick off each item before you submit:

☐	Written report (PDF or Word) with all 5 sub-sections complete
☐	Minimum 6 credible sources cited (NASA, ESA, science journals preferred)
☐	All images in report have captions and are relevant to the content
☐	Reflection section is personal, genuine, and 150–200 words
☐	Website folder submitted with index.html as the entry point
☐	All 6 website pages/sections present, complete, and functional
☐	CSS is in a separate stylesheet file
☐	At least 1 JavaScript feature implemented and working
☐	All images on website have descriptive alt text
☐	README.txt included explaining how to open the site
☐	Screenshots of every website section submitted as evidence

"Somewhere, something incredible is waiting to be known."

— Sharon Begley